Examples:

Standard Form	Scientific Notation	Standard Form	Scientific Notation
0.00123	1.23×10^{-3}	123	1.23×10^{2}
0.0123	1.23×10^{-2}	1230	1.23×10^{3}
0.123	1.23×10^{-1}	12300	1.23×10^{4}
1.23	1.23×10^{0}	123000	1.23×10^{5}
12.3	1.23×10^{1}	1230000	1.23×10^{6}

Some more examples:

(a) The diameter of the earth is 12756000 miles. This can be easily written in scientific form as 1.2756×10^{7} miles.

(b) The volume of Jupiter is about 143300000000000 km^3. This can be easily written in scientific form as 1.433×10^{14} km^3.

(c) The size of a bacterium is 0.00000085 mm. This can be easily written in scientific form as 8.5×10^{-7} mm.



Note

1. The positive exponent in 1.3×10^{12} indicates that it is a large number.
2. The negative exponent in 7.89×10^{-21} indicates that it is a small number.

Example 1.39

Combine the scientific notations: (i) $\left(7 \times 10^{2}\right)\left(5.2 \times 10^{7}\right)$ (ii) $\left(3.7 \times 10^{-5}\right)\left(2 \times 10^{-3}\right)$

Solution:

(i) $\left(7 \times 10^{2}\right)\left(5.2 \times 10^{7}\right) = 36.4 \times 10^{9} = 3.64 \times 10^{10}$

(ii) $\left(3.7 \times 10^{-5}\right)\left(2 \times 10^{-3}\right) = 7.4 \times 10^{-8}$

Example 1.40

Write the following scientific notations in standard form:

(i) 2.27×10^{-4} (ii) Light travels at 1.86×10^{5} miles per second.

Solution:

(i) $2.27 \times 10^{-4} = 0.000227$.

(ii) Light travels at 1.86×10^{5} miles per second = 186000 miles per second



Try these

1. Write in standard form: Mass of planet Uranus is 8.68×10^{25} *kg*.
2. Write in scientific notation: (i) 0.000012005 (ii) 4312.345 (iii) 0.10524 (iv) The distance between the Sun and the planet Saturn 1.4335×10^{12} miles.

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